

# Ziang Li

Master's student in mathematics  
University of Regensburg | Regensburg, Germany  
Ziang.Li@stud.uni-regensburg.de

## Education

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Oct 2025 - Present | <b>University of Regensburg</b><br>Master's studies in mathematics; Regensburg, Germany<br><b>Master thesis in progress:</b> An Isomorphism Between the Blow-Up of the Deligne–Mumford Moduli Space and the Decorated Deligne–Mumford Moduli Space.<br><b>Supervisor(s):</b> Prof. Stefan Friedl and Prof. Bernd Ammann<br><b>Selected coursework:</b><br><i>Grades in Germany are on a scale from 1.0 to 5.0; 5.0 is a failing grade.</i><br>Twisted homology and Reidemeister torsion (1.3)<br>Simple homotopy theory and manifold topology (1.0)<br>Seminar on Floer homology, Part II (1.0) | Algebraic Topology I (1.3)<br>Algebraic Topology II (1.0)<br>Characteristic classes (1.3) |
|                    | <b>Xi'an Jiaotong University</b><br>Bachelor's degree in mathematics; Xi'an, China<br><b>Thesis:</b> Quasi-isometry Groups of $\mathbb{R}^n$ .<br><b>Supervisor:</b> Qiang Zhang.<br><b>Selected coursework:</b><br>Riemannian Geometry (100)<br>Graph Theory and Combinatorics (92)<br>Mathematical Analysis Seminar (99)                                                                                                                                                                                                                                                                      | Functional Analysis (90)<br>Algebra Seminar (92)<br>Differentiable Manifolds (90)         |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                           |

## Research interests

Geometric topology, especially knot theory and the topology of 3- and 4-manifolds.

## Academic experience

### Seminar talks

|          |                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jul 2026 | <b>Proof of the s-cobordism theorem</b><br>Simple homotopy theory and manifold topology seminar, University of Regensburg<br>Organizer: Dr. Marco Volpe. Report                            |
| Jul 2026 | <b>A surgery formula for Seiberg-Witten invariants</b><br>Seminar on spin geometry, University of Regensburg<br>Organizer: Prof. Bernd Ammann. Report                                      |
| Dec 2025 | <b>Proof of the Arnold conjecture</b><br>Seminar on Floer homology, Part II, University of Regensburg<br>Organizer: Prof. Bernd Ammann. Report                                             |
| Nov 2025 | <b>An introduction to Seiberg-Witten theory and its applications to 4-manifolds</b><br>Surfaces in 4-manifolds seminar, University of Regensburg<br>Organizer: Prof. Stefan Friedl. Report |

## Research notes

|          |                                                                                                                                                                                                                                                               |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jul 2026 | <b>Direct proof of equivalence of two definitions of Alexander polynomials</b><br>Supervised by: Prof. Stefan Friedl.<br>I gave a direct proof that the definitions of the Alexander polynomial via the Burau representation and Fox calculus are equivalent. |
| Sep 2026 | <b>Master thesis in progress: An Isomorphism Between the Blow-Up of the Deligne–Mumford Moduli Space and the Decorated Deligne–Mumford Moduli Space.</b><br>Supervised by: Prof. Bernd Ammann.<br>Tba                                                         |

## Conferences, summer schools, and workshops

---

|                              |                                                                                                                                                                                  |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 Jun 2026 -<br>12 Jun 2026  | <b>Masterclass: Seiberg-Witten Invariants</b><br>University of Copenhagen; Copenhagen, Denmark. <a href="#">Program</a>                                                          |
| 18 May 2026 -<br>22 May 2026 | <b>Spring school "Khovanov and Floer homotopy"</b><br>Nantes University; Nantes, France. <a href="#">Program</a>                                                                 |
| 27 Dec 2025 -<br>28 Dec 2025 | <b>ALGANT Alumni in China</b><br>University of Science and Technology of China; Hefei, China. <a href="#">Program</a>                                                            |
| 8 Dec 2025 -<br>12 Dec 2025  | <b>Workshop on surfaces in 4-manifolds</b><br>University of Regensburg; Regensburg, Germany. <a href="#">Program</a>                                                             |
| 1 Jul 2025 -<br>11 Jul 2025  | <b>USTC's 2025 Summer School in Geometry and Analysis</b><br>University of Science and Technology of China; Hefei, China. <a href="#">Program</a>                                |
| 22 Aug 2024 -<br>28 Aug 2024 | <b>SUSTech's 2024 Number Theory and Arithmetic Geometry Conference</b><br>Southern University of Science and Technology; Shenzhen, China. <a href="#">Program</a>                |
| 29 Jul 2024 -<br>9 Aug 2024  | <b>BICMR's 2024 Summer School in Differential Geometry</b><br>Beijing International Center for Mathematical Research, Peking University; Beijing, China. <a href="#">Program</a> |
| 1 Jul 2024 -<br>9 Jul 2024   | <b>YMSC's 2024 Summer School on Frontiers in Topology</b><br>Yau Mathematical Sciences Center, Tsinghua University; Beijing, China. <a href="#">Program</a>                      |

## Student-led reading seminars

---

|                     |                                                                            |
|---------------------|----------------------------------------------------------------------------|
| Sep 2024 - Jun 2025 | Symplectic geometry (Organizer and main speaker)                           |
| Mar 2024 - Jun 2024 | Characteristic classes (Organizer and main speaker)                        |
| Dec 2023 - Mar 2024 | Morse theory (Organizer and main speaker)                                  |
| Sep 2023 - Jun 2024 | Category theory and homological algebra (Participant)                      |
| Mar 2022 - Jan 2023 | Algebraic topology (Participant and speaker)                               |
| Feb 2022 - Jun 2022 | Riemannian geometry (Organizer and main speaker)                           |
| Sep 2021 - Jan 2022 | Differential manifolds and de Rham cohomology (Organizer and main speaker) |

Last updated: October 2026.